

BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/BEE 4750/hw2-regina`
Installed InlineStrings          v1.4.5
Installed Fontconfig_jll         v2.17.1+0
Installed Libmount_jll           v2.41.1+0
Installed JpegTurbo_jll          v3.1.3+0
Installed InvertedIndices         v1.3.1
Installed Accessors              v0.1.42
Installed Roots                  v2.2.10
Installed DataFrames             v1.8.0
Installed SentinelArrays         v1.4.8
Installed DataStructures         v0.19.1
Installed StatsBase              v0.34.6
Installed Plots                   v1.40.19
Installed WeakRefStrings         v1.4.2
Installed WorkerUtilities        v1.6.1
Installed Expat_jll              v2.7.1+0
Installed TableTraits            v1.0.1
Installed PrettyTables           v3.0.8
Installed Ghostscript_jll        v9.55.1+0
Installed ConstructionBase       v1.6.0
Installed PooledArrays           v1.4.3
Installed CompositionsBase       v0.1.2
Installed DataValueInterfaces    v1.0.0
Installed InverseFunctions       v0.1.17
Installed CommonSolve            v0.2.4
Installed Libuuid_jll            v2.41.1+0
Installed OpenSSL_jll            v3.5.2+0
Installed Latexify                v0.16.10
Installed Tables                 v1.12.1
Installed Glib_jll               v2.86.0+0
Installed Crayons                 v4.1.1
Installed StringManipulation     v0.4.1
Installed SortingAlgorithms      v1.2.2
Installed CSV                     v0.10.15
Installed IteratorInterfaceExtensions v1.0.0
Installed FilePathsBase          v0.9.24
Precompiling project...
 264.5 ms  IteratorInterfaceExtensions
 198.8 ms  DataValueInterfaces
 210.8 ms  CommonSolve
 263.5 ms  WorkerUtilities
 262.9 ms  InverseFunctions
 210.7 ms  CompositionsBase

```

240.7 ms	InvertedIndices
360.4 ms	InlineStrings
268.4 ms	ConstructionBase
318.7 ms	PooledArrays
334.1 ms	OpenSSL_jll
612.1 ms	Crayons
317.5 ms	Graphite2_jll
281.1 ms	Libmount_jll
290.4 ms	EpollShim_jll
743.0 ms	SentinelArrays
309.7 ms	LLVMOpenMP_jll
295.8 ms	Bzip2_jll
283.8 ms	Xorg_libICE_jll
1111.7 ms	DataStructures
283.9 ms	Xorg_libXau_jll
321.0 ms	libpng_jll
299.0 ms	libfdk_aac_jll
334.1 ms	LAME_jll
315.3 ms	LERC_jll
314.9 ms	fzf_jll
347.9 ms	JpegTurbo_jll
303.4 ms	Ogg_jll
345.9 ms	XZ_jll
275.7 ms	mtdev_jll
286.7 ms	Xorg_libXdmcp_jll
318.0 ms	x265_jll
310.7 ms	libaom_jll
327.7 ms	x264_jll
311.5 ms	Zstd_jll
313.7 ms	Expat_jll
292.6 ms	Opus_jll
311.9 ms	LZO_jll
288.5 ms	Xorg_xtrans_jll
283.0 ms	libevdev_jll
318.6 ms	Libiconv_jll
282.7 ms	eudev_jll
274.3 ms	Libuuid_jll
307.0 ms	Libffi_jll
320.2 ms	FriBidi_jll
472.9 ms	FilePathsBase
529.7 ms	TableTraits
807.1 ms	RecipesBase
351.2 ms	InverseFunctions → InverseFunctionsDatesExt

321.8 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
456.1 ms	InverseFunctions → InverseFunctionsTestExt
973.7 ms	StringManipulation
317.0 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
388.2 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
438.1 ms	Unitful → InverseFunctionsUnitfulExt
501.4 ms	InlineStrings → ParsersExt
316.8 ms	Unitful → ConstructionBaseUnitfulExt
256.1 ms	Xorg_libSM_jll
495.5 ms	Pixman_jll
505.2 ms	FreeType2_jll
438.4 ms	SortingAlgorithms
2310.2 ms	ColorSchemes
705.5 ms	JLFFzf
268.7 ms	Xorg_libxcb_jll
439.1 ms	libvorbis_jll
1474.8 ms	OpenSSL
299.4 ms	DBus_jll
1079.5 ms	Ghostscript_jll
274.7 ms	libinput_jll
267.5 ms	Wayland_jll
597.4 ms	GettextRuntime_jll
782.6 ms	Libtiff_jll
368.1 ms	FilePathsBase → FilePathsBaseMmapExt
590.1 ms	FilePathsBase → FilePathsBaseTestExt
666.6 ms	Tables
584.4 ms	WeakRefStrings
221.2 ms	Xorg_xcb_util_jll
234.3 ms	Xorg_libX11_jll
1415.5 ms	Fontconfig_jll
1579.0 ms	Accessors
1609.0 ms	StatsBase
890.5 ms	Glib_jll
2011.1 ms	Latexify
284.4 ms	Xorg_xcb_util_image_jll
314.7 ms	Xorg_xcb_util_keysyms_jll
317.2 ms	Xorg_xcb_util_renderutil_jll
354.5 ms	Xorg_xcb_util_wm_jll
297.1 ms	Xorg_libXrender_jll
301.6 ms	Xorg_libXext_jll
281.2 ms	Xorg_libXfixes_jll
337.6 ms	Xorg_libxkbfile_jll
5606.4 ms	PlotUtils

434.2 ms	Accessors → LinearAlgebraExt
363.7 ms	Accessors → TestExt
355.4 ms	Accessors → UnitfulExt
445.0 ms	Latexify → SparseArraysExt
778.4 ms	UnitfulLatexify
314.2 ms	Xorg_xcb_util_cursor_jll
319.8 ms	Xorg_libXinerama_jll
293.9 ms	Xorg_libXrandr_jll
296.8 ms	Libglvnd_jll
277.2 ms	Xorg_libXcursor_jll
312.1 ms	Xorg_libXi_jll
329.1 ms	Xorg_xkbcomp_jll
1714.6 ms	Cairo_jll
1749.4 ms	PlotThemes
10042.7 ms	HTTP
8886.8 ms	CSV
402.1 ms	Xorg_xkeyboard_config_jll
326.3 ms	xkbcommon_jll
2479.0 ms	RecipesPipeline
723.3 ms	HarfBuzz_jll
335.2 ms	Vulkan_Loader_jll
1846.0 ms	Roots
718.1 ms	libass_jll
753.2 ms	Pango_jll
548.2 ms	Roots → RootsUnitfulExt
309.4 ms	libdecor_jll
269.0 ms	GLFW_jll
2423.7 ms	Qt6Base_jll
1879.7 ms	FFMPEG_jll
805.6 ms	FFMPEG
872.9 ms	Qt6ShaderTools_jll
1192.2 ms	GR_jll
1067.6 ms	Qt6Declarative_jll
247.7 ms	Qt6Wayland_jll
1860.1 ms	GR
17196.6 ms	PrettyTables
20905.9 ms	DataFrames
1172.0 ms	Latexify → DataFramesExt
28423.5 ms	Plots
2245.1 ms	Plots → UnitfulExt

132 dependencies successfully precompiled in 56 seconds. 87 already precompiled.

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

[Info: Precompiling IJuliaExt [2f4121a4-3b3a-5ce6-9c5e-1f2673ce168a] (cache misses: wrong d

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```
"""
      (+)
Atmospheric CO2      Global Mean Temperature
                        |
                        (-)
                        |
                        Ocean CO2
(-)      -
"""
```

```
"      (+)\nAtmospheric CO2      Global Mean Temperature\n
```

Problem 1 SOLUTION:

Henry's Law says that solubility of CO₂ in seawater decreases as temperature increases. In other words, warm water holds less CO₂. Because of this, the arrow pointing from global mean temperature to the concentration of CO₂ in the ocean is a negative (-) sign and this is a dampening relationship. As temperature increases, the ocean's water warms and CO₂ concentrations decrease.

When concentrations of CO₂ in the ocean increase, the concentration of CO₂ in the atmosphere decrease since the ocean is acting as a carbon sink and taking out CO₂ from the atmosphere. Because of this, the sign is negative (-) and this is a dampening relationship.

Finally, when CO₂ concentrations in the atmosphere increases, the greenhouse effect is magnified since the CO₂ is now absorbing heat and re-emitting it. As a result, the global mean temperature to increase. This is an amplifying relationship and the arrow's sign is positive (+).

This suggests that the overall feedback between temperature and the ocean carbon cycle is an amplifying (or positive) feedback because an increase in atmospheric CO₂, increases temperature, which creates less CO₂ in the ocean, which means less CO₂ is being absorbed by the ocean and there is more of it in the atmosphere, which finally leads to even more warming.

Tip

Think about Henry's law for CO₂.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system is in compliance with a regulatory limit of $2.3 \text{ kg}/(1000\text{m}^3)$. You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in $^{\circ}\text{C}$);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/^{\circ}\text{C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/^{\circ}\text{C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

References

List any external references consulted, including classmates.